

Spam Detection

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1 PROBLEM DESCRIPTION

The dataset consists of information from 4601 e-mail messages, in a study to screen e-mail for spam. For 3101 of these e-mails you know whether they are spam [response = 2] or non-spam [response = 1], and you have additional input variables describing several features of each e-mail. For the other 1500 e-mails, you have only information on the inputs.

Your goal is to construct a classifier which has good performance in labeling the remaining 1500 e-mails as spam or non-spam. **Note** that in classifying as spam an e-mail which is actually non-spam you pay a cost of 5. Classifying as non-spam e-mails which are actually spam is less dangerous and hence you pay a cost of 1.

There are 57 input variables which are described below.

- 48 quantitative predictors measuring the percentage of words in the e-mail that match a given word. Examples include business, address, internet, etc ...
- 6 quantitative predictors measuring the percentage of specific characters/symbols found in the e-mail. These are charSemicolon, charRoundbracket, charSquarebracket, charExclamation, charDollar and charHash.
- Average length of uninterrupted sequences of capital letters [capitalAve]. Length of the longest uninterrupted sequence of capital letters [capitalLong]. Sum of the length of uninterrupted sequences of capital letters [CapitaTotal].